

CBSE Class 12 Physics Question Paper 2024

Set 1 (55/1/1) Solutions

SECTION A

1. A thin plastic rod is bent into a circular ring of radius R . It is uniformly charged with charge density λ . The magnitude of the electric field at its centre is :

(A) $\frac{\lambda}{2\epsilon_0 R}$

(B) Zero

(C) $\frac{\lambda}{4\pi\epsilon_0 R}$

(D) $\frac{\lambda}{4\epsilon_0 R}$

Ans: (B) Zero

2. Ten capacitors, each of capacitance $1\mu\text{F}$, are connected in parallel to a source of 100 V. The total energy stored in the system is equal to :

(A) 10^{-2} J

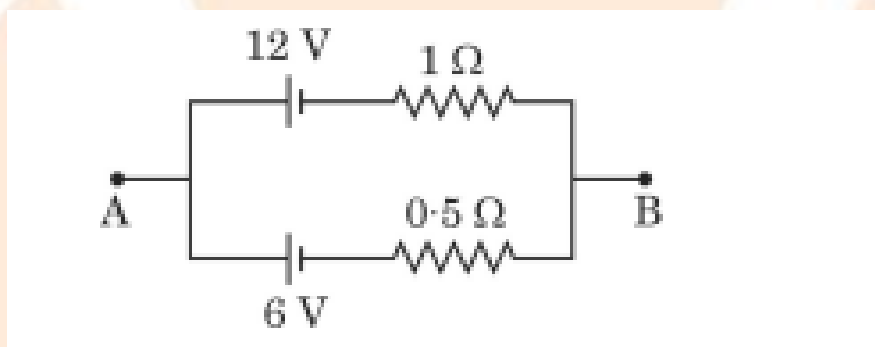
(B) 10^{-3} J

(C) $0.5 \times 10^{-3} \text{ J}$

(D) $5.0 \times 10^{-2} \text{ J}$

Ans: (D) $5 \cdot 0 \times 10^{-2} \text{ J}$

3. Consider the circuit shown in the figure. The potential difference between points A and B is:



(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

Ans: (A) 6 V

4. A loop carrying a current I clockwise is placed in x - y plane, in a uniform magnetic field directed along z -axis. The tendency of the loop will be to :

(A) move along x -axis

(B) move along y -axis

(C) shrink

(D) expand

Ans: (C) shrink

5. A 10 cm long wire lies along y-axis. It carries a current of 1.0 A in positive y-direction. A magnetic field $\vec{B} = (5\text{mT})\hat{j} - (8\text{mT})\hat{k}$ exists in the region. The force on the wire is :

(A) $(0.8\text{mN})\hat{i}$

(B) $-(0.8\text{mN})\hat{i}$

(C) $(80\text{mN})\hat{i}$

(D) $-(80\text{mN})\hat{i}$

Ans: (D) $-(80\text{mN})\hat{i}$

6. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to 1 A. If the current through the galvanometer is 0.1% of 1 A, the resistance of the ammeter is:

(A) $\frac{G}{999}$

(B) $\frac{G}{1000}$

(C) $1000 G \Omega$

(D) $1001 G \Omega$

Answer: (A) $\frac{G}{999}$

7. The reactance of a capacitor of capacitance C connected to an ac source of frequency ω is 'X'. If the capacitance of the capacitor is doubled and the frequency of the source is tripled, the reactance will become:

- (A) $\frac{X}{6}$
- (B) $6X$
- (C) $\frac{2X}{3}$
- (D) $\frac{3X}{2}$

Answer: (A) $\frac{X}{6}$

8. In the four regions, I, II, III, and IV, the electric fields are described as:

- (A) Region I: $E_x = E_0 \sin(kz - \omega t)$
- (B) Region II: $E_x = E_0$
- (C) Region III: $E_x = E_0 \sin kz$
- (D) Region IV: $E_x = E_0 \cos kz$

The displacement current will exist in the region:

- (A) I
- (B) IV
- (C) II
- (D) III

Answer: (A) I

9. The transition of an electron that gives rise to the formation of the second spectral line of the Balmer series in the spectrum of the hydrogen atom corresponds to:

(A) $n_f = 2$ and $n_i = 3$

(B) $n_f = 3$ and $n_i = 4$

(C) $n_f = 2$ and $n_i = 4$

(D) $n_f = 2$ and $n_i = 5$

Answer: (C) $n_f = 2$ and $n_i = 4$

10. Ge is doped with As. Due to doping,

(A) the structure of Ge lattice is distorted.

(B) the number of conduction electrons increases.

(C) the number of holes increases.

(D) the number of conduction electrons decreases.

Answer: (B) the number of conduction electrons increases.

11. Two beams, A and B whose photon energies are 3.3 eV and 11.3 eV respectively, illuminate a metallic surface (work function 2.3 eV) successively. The ratio of the maximum speed of electrons emitted due to beam A to that due to beam B is:

(A) 3

(B) 9

(C) $\frac{1}{3}$

(D) $\frac{1}{9}$

Answer: (D) 3

12. The waves associated with a moving electron and a moving proton have the same wavelength. It implies that they have the same:

(A) momentum (B) angular momentum (C) speed (D) energy

Ans: (A) momentum

Questions number 13 to 16 are Assertion (A) and Reason (R) type questions. Two statements are given one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

13. Assertion (A) : In photoelectric effect, the kinetic energy of the emitted photoelectrons increases with increase in the intensity of the incident light.

Reason (R) : Photoelectric current depends on the wavelength of the incident light.

Ans: (D) Assertion (A) is false and Reason (R) is also false.

14. Assertion (A): The mutual inductance between two coils is maximum when the coils are wound on each other.

Reason (R): The flux linkage between two coils is maximum when they are wound on each other.

Ans: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

15. Assertion (A) : Two long parallel wires, freely suspended and connected in series to a battery, move apart.

Reason (R) : Two wires carrying current in opposite directions repel each other.

Ans: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

16. Assertion (A) : Plane and convex mirrors cannot produce real images under any circumstance.

Reason (R) : A virtual image cannot serve as an object to produce a real image.

Ans: (C) Assertion (A) is true, but Reason (R) is false.

SECTION B

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20° C. Take 20°C as the reference temperature and temperature coefficient of resistance of silver at 20°C = $4.0 \times 10^{-3} \text{ K}^{-1}$.

2

Ans:

To find the temperature T where the resistance R is twice its resistance at 20°C:

$$R_T = R_{20} [1 + \alpha(T - 20)]$$

Given that $R_T = 2R_{20}$ and $\alpha = 4.0 \times 10^{-3} \text{ K}^{-1}$.

$$2R_{20} = R_{20} [1 + 4.0 \times 10^{-3}(T - 20)]$$

$$2 = 1 + 4.0 \times 10^{-3}(T - 20)$$

$$1 = 4.0 \times 10^{-3}(T - 20)$$

$$T - 20 = \frac{1}{4.0 \times 10^{-3}}$$

$$T - 20 = 250$$

$$T = 270^\circ\text{C}$$

This is the required temperature.

18. (a) Monochromatic light of frequency $5.0 \times 10^{14} \text{ Hz}$ passes from air into a medium of refractive index 1.5. Find the wavelength of the light (i) reflected, and (ii) refracted at the interface of the two media.

Ans:

(i) The wavelength of the reflected light remains the same as in air.

$$\lambda_{\text{air}} = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{5.0 \times 10^{14} \text{ Hz}} = 6 \times 10^{-7} \text{ m} = 600 \text{ nm}$$

(ii) The wavelength of the refracted light in the medium with refractive index $n=1.5$:

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{air}}}{n} = \frac{600 \text{ nm}}{1.5} = 400 \text{ nm}$$

Required values are: (i) 600 nm, (ii) 400 nm

OR

(b) A plano-convex lens of focal length 16 cm is made of a material of refractive index 1.4. Calculate the radius of the curved surface of the lens. **2**

Ans:

Using the lens maker's formula:

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For a plano-convex lens, $R_1 = R$ and $R_2 = \infty$.

$$\frac{1}{16} = (1.4-1) \left(\frac{1}{R} - 0 \right)$$

$$\frac{1}{16} = 0.4 \times \frac{1}{R}$$

$$R = 0.4 \times 16 = 6.4 \text{ cm}$$

Radius of the curved surface is 6.4 cm

19. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm . Find the (i) position of the image formed and (ii) magnification of the image. 2

Ans: (i) Using the mirror formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

The focal length $f = \frac{R}{2} = \frac{40}{2} = 20 \text{ cm}$, $u = -30 \text{ cm}$

$$\frac{1}{20} = \frac{1}{v} + \frac{1}{-30}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{30} = \frac{3+2}{60} = \frac{5}{60} = \frac{1}{12}$$

$$v = 12 \text{ cm}$$

(ii) Magnification $m = -\frac{v}{u}$

$$m = -\frac{12}{-30} = \frac{12}{30} = 0.4$$

Therefore, (i) Position of the image is 12 cm in front of the mirror, (ii) Magnification is 0.4

20. Consider a neutron (mass m) of kinetic energy E and a photon of the same energy. Let λ_n and λ_p be the de Broglie wavelength of neutron and the wavelength of photon respectively. Obtain an expression for $\frac{\lambda_n}{\lambda_p}$. 2

Ans: For the neutron:

$$E = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mE}$$

$$\lambda_n = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

For the photon:

$$E = \frac{hc}{\lambda_p}$$

$$\lambda_p = \frac{hc}{E}$$

Therefore,

$$\frac{\lambda_n}{\lambda_p} = \frac{\frac{h}{\sqrt{2mE}}}{\frac{hc}{E}} = \frac{E}{h} \cdot \frac{h}{\sqrt{2mE}} \cdot \frac{1}{c} = \frac{E}{\sqrt{2mEc}} = \frac{\sqrt{E}}{\sqrt{2mc}}$$

The required ratio of $\frac{\lambda_n}{\lambda_p} = \frac{\sqrt{E}}{\sqrt{2mc}}$.

21. Plot a graph showing the variation of current with voltage for the material GaAs. On the graph, mark the region where: **2**

(a) resistance is negative, and

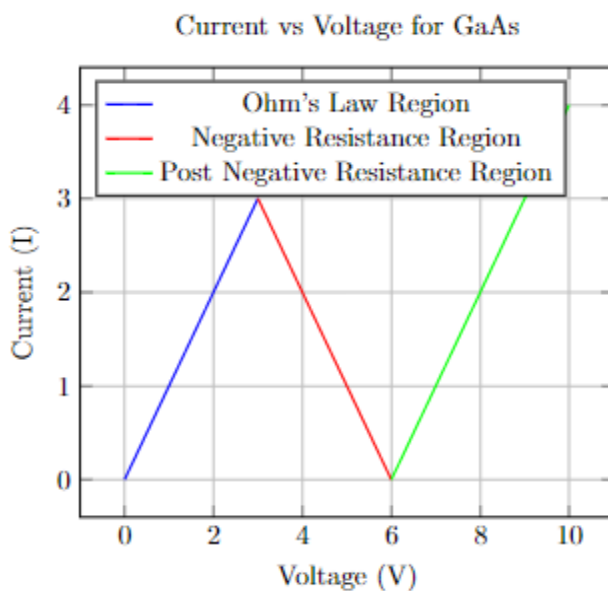
(b) Ohm's law is obeyed.

Ans:

(a) The region where resistance is negative will show a decrease in current with an increase in voltage.

(b) The region where Ohm's law is obeyed will show a linear relationship between current and voltage.

The graph showing the variation of current with voltage for the material GaAs, highlighting the regions where resistance is negative and where Ohm's law is obeyed:



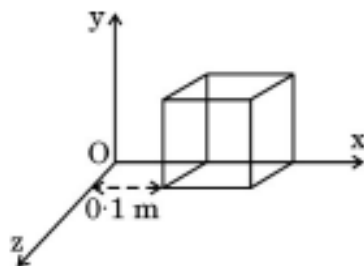
SECTION C

22. A cube of side 0–1m is placed, as shown in the figure, in a region where electric field $\vec{E} = 500 \times \hat{i}$ exists. Here x is in meters and E in N^{-1} . Calculate:

3

(a) the flux passing through the cube, and

(b) the charge within the cube.



Ans:

(a) Flux through the cube:

The electric flux Φ_E through a surface is given by:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

For a cube in a non-uniform electric field $\vec{E} = 500x\hat{i}$:

$$\Phi_E = \int_0^{0.1} E(x) \cdot A dx = \int_0^{0.1} 500x \cdot (0.1)^2 dx$$

$$\Phi_E = 500 \times 0.01 \times \int_0^{0.1} x dx$$

$$\Phi_E = 5 \times \left[\frac{x^2}{2} \right]_0^{0.1} = 5 \times \frac{(0.1)^2}{2} = 5 \times 0.005 = 0.025 \text{ Nm}^2/\text{C}$$

(b) Charge within the cube:

Using Gauss's law:

$$\Phi_E = \frac{q_{\text{in}}}{\epsilon_0}$$

$$q_{\text{in}} = \Phi_E \times \epsilon_0 = 0.025 \times 8.85 \times 10^{-12} \text{ C} = 2.21 \times 10^{-13} \text{ C}$$

23. (a) Define 'current density'. Is it a scalar or a vector? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m , charge $-e$) per unit volume in the conductor and τ its relaxation time, show that the current

density $\vec{j} = \alpha \vec{E}$, where $\alpha = \left(\frac{ne^2}{m} \right) \tau$.

3

Ans: Current Density Definition: Current density $\vec{J}$ is the amount of electric current flowing per unit area of a cross-section. It is a vector quantity.

Derivation:

$\vec{J} = ne\vec{v}_d$, where $\vec{v}_d$ is the drift velocity, given by:

$$\vec{v}_d = \frac{e\vec{E}\tau}{m}$$

Therefore:

$$\vec{J} = ne \left(\frac{e\vec{E}\tau}{m} \right) = \left(\frac{ne^2\tau}{m} \right) \vec{E}$$

Hence, $\alpha = \frac{ne^2\tau}{m}$

OR

(b) What is a Wheatstone bridge? Obtain the necessary conditions under which the Wheatstone bridge is balanced.

3

Ans: Wheatstone Bridge:

A Wheatstone bridge is an electrical circuit used to measure an unknown resistance by balancing two legs of a bridge circuit.

Balanced Condition:

The bridge is balanced when:

$$\frac{R_1}{R_2} = \frac{R_3}{R_x} \text{ where, } R_1, R_2, R_3 \text{ are known resistances and } R_x \text{ is the unknown resistance.}$$

24. A proton with kinetic energy 1.3384×10^{-14} J moving horizontally from north to south, enters a uniform magnetic field B of 2.0 mT directed eastward. Calculate:

3

- (a) the speed of the proton**
- (b) the magnitude of acceleration of the proton**
- (c) the radius of the path traced by the proton**

[Take (q/m) for proton $= 1.0 \times 10^8 \text{ C/kg}$]

Ans:

(a) Speed of the proton:

$$K = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times 1.3384 \times 10^{-14}}{1.67 \times 10^{-27}}} = \sqrt{1.602 \times 10^{13}} = 1.26 \times 10^6 \text{ m/s}$$

(b) Magnitude of acceleration:

The proton undergoes circular motion due to the magnetic field:

$$a = \frac{v^2}{r}$$

$$\text{But, } r = \frac{mv}{eB}$$

(c) Radius of the path:

$$r = \frac{1.67 \times 10^{-27} \times 1.26 \times 10^6}{1.6 \times 10^{-19} \times 2.0 \times 10^{-3}} = \frac{2.1042 \times 10^{-21}}{3.2 \times 10^{-22}} = 6.57 \text{ m}$$

Calculate 'a':

$$a = \frac{(1.26 \times 10^6)^2}{6.57} = \frac{1.5876 \times 10^{12}}{6.57} = 2.42 \times 10^{11} \text{ m/s}^2$$

This gives:

- (a) Speed of the proton: $1.26 \times 10^6 \text{ m/s}$
- (b) Magnitude of acceleration: $2.42 \times 10^{11} \text{ m/s}^2$
- (c) Radius of the path: 6.57 m .

25. An inductor, a capacitor and a resistor are connected in series with an ac source $v = v_m \sin \omega t$. Derive an expression for the average power dissipated in the circuit. Also obtain the expression for the resonant frequency of the circuit. **3**

Ans: Average Power Dissipated:

The impedance Z of the series RLC circuit is:

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

The current in the circuit is:

$$I = \frac{v_m}{Z} \sin(\omega t - \phi)$$

The average power P dissipated is given by:

$$P = I_{\text{rms}}^2 R = \left(\frac{v_m}{\sqrt{2}Z}\right)^2 R = \frac{v_m^2}{2} \cdot \frac{R}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Resonant Frequency:

Resonance occurs when the reactive components cancel each other out:

$$\omega L = \frac{1}{\omega C}$$

The resonant frequency ω_0 is:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Therefore,

- Average power dissipated is $P = \frac{V_m^2}{2} \cdot \frac{R}{R^2 + (\omega L - \frac{1}{\omega C})^2}$
- Resonant frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$

26. (a) "The wavelength of the electromagnetic wave is often correlated with the characteristic size of the system that radiates." Give two examples to justify this statement.

(b) (i) Long distance radio broadcasts use short-wave bands. Why?

Ans: Short waves can reflect off the ionosphere, allowing them to travel long distances by bouncing between the Earth and the ionosphere.

(ii) Optical and radio telescopes are built on the ground, but X-ray astronomy is possible only from satellites orbiting the Earth. Why?

3

Ans: X-rays are absorbed by the Earth's atmosphere and do not reach the ground, necessitating the placement of X-ray telescopes in orbit.

27. Write the drawbacks of Rutherford's atomic model. How did Bohr remove them? Show that different orbits in Bohr's atom are not equally spaced. **3**

Ans:

Drawbacks of Rutherford's Model:

- Could not explain the stability of the atom.
- Could not explain the discrete spectra of atoms.

Bohr's Improvements:

- Introduced quantized orbits where electrons can only occupy specific energy levels.
- Explained that electrons do not radiate energy while in these stationary orbits, thus ensuring atomic stability.

Spacing of Orbits in Bohr's Model:

- The energy of an electron in the n-th orbit is given by:

$$E_n = -\frac{13.6\text{eV}}{n^2}$$

- The energy difference between orbits n and n+1 is:

$$\Delta E = E_{n+1} - E_n = -\frac{13.6\text{eV}}{(n+1)^2} + \frac{13.6\text{eV}}{n^2}$$

This shows that the energy levels are not equally spaced.

28. (a) State any two properties of a nucleus.

Ans:

- Very small in size compared to the atom.
- Contains most of the atom's mass, which is concentrated in the nucleus

(b) Why is the density of a nucleus much more than that of an atom?

Ans: The nucleus contains protons and neutrons packed tightly together, while the atom consists mostly of empty space with electrons orbiting at a distance.

(c) Show that the density of the nuclear matter is the same for all nuclei.

3

The nuclear radius

$R = R_0 A^{1/3}$, where R_0 is a constant and A is the mass number. The volume V of the nucleus is:

$$V = \frac{4}{3} \pi R^3 = \frac{4}{3} \pi (R_0 A^{1/3})^3 = \frac{4}{3} \pi R_0^3 A$$

The density ρ is:

$$\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{m_p A}{\frac{4}{3} \pi R_0^3 A} = \frac{m_p}{\frac{4}{3} \pi R_0^3}$$

Since m_p and R_0 are constants, the density is the same for all nuclei.

SECTION D

Questions number 29 and 30 are case study-based questions. Read the following paragraphs and answer the questions that follow.

29. A lens is a transparent medium bounded by two surfaces, with one or both surfaces being spherical. The focal length of a lens is determined by the radii of curvature of its two surfaces and the refractive index of its medium with respect to that of the surrounding medium. The power of a lens is reciprocal of its focal length. If a number of lenses are kept in contact, the power of the combination is the algebraic sum of the powers of the individual lenses.

(i) A double-convex lens, with each face having same radius of curvature R , is made of glass of refractive index n . Its power is :

1

(A) $\frac{2(n-1)}{R}$

(B) $\frac{(2n-1)}{R}$

(C) $\frac{(n-1)}{2R}$

(D) $\frac{(2n-1)}{2R}$

Ans: (A) $\frac{2(n-1)}{R}$

(ii) A double-convex lens of power P , with each face having same radius of curvature, is cut into two equal parts perpendicular to its principal axis. The power of one part of the lens will be :

1

(A) $2P$

(B) P

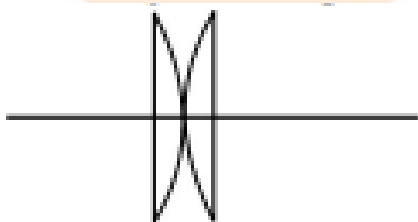
(C) 4 P

(D) $\frac{P}{2}$

Ans: (B) P

(iii) The above two parts are kept in contact with each other as shown in the figure. The power of the combination will be :

1



(A) $\frac{P}{2}$

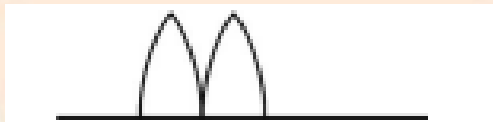
(B) P

(C) 2 P

(D) $\frac{P}{4}$

Ans: (C) 2 P

(iv) (a) A double-convex lens of power P , with each face having same radius of curvature, is cut along its principal axis. The two parts are arranged as shown in the figure. The power of the combination will be: 1



(A) Zero

(B) P

(C) $2P$

(D) $\frac{P}{2}$

Ans: (B) P

OR

(b) Two convex lenses of focal lengths 60 cm and 20 cm are held coaxially in contact with each other. The power of the combination is : 1

(A) 6.6D

(B) 15 D

(C) $\frac{1}{15}$ D

(D) $\frac{1}{80}$ D

Ans: (A) 6.67 D

30. Junction Diode as a Rectifier :

The process of conversion of an ac voltage into a dc voltage is called rectification and the device which performs this conversion is called a rectifier. The characteristics of a p-n junction diode reveal that when a p-n junction diode is forward biased, it offers a low resistance and when it is reverse biased, it offers a high resistance. Hence, a p-n junction diode conducts only when it is forward biased. This property of a p-n junction diode makes it suitable for its use as a rectifier.

Thus, when an ac voltage is applied across a p-n junction, it conducts only during those alternate half cycles for which it is forward biased. A rectifier which rectifies only half cycle of an ac voltage is called a half-wave rectifier and one that rectifies both the half cycles is known as a full-wave rectifier.

(i) The root mean square value of an alternating voltage applied to a full-wave rectifier is $\frac{V_0}{\sqrt{2}}$. Then the root mean square value of the rectified output voltage is:

- (A) $\frac{V_0}{\sqrt{2}}$ (B) $\frac{V_0^2}{\sqrt{2}}$ (C) $\frac{2V_0}{\sqrt{2}}$ (D) $\frac{V_0}{2\sqrt{2}}$

Ans: (A) $\frac{V_0}{\sqrt{2}}$

(ii) In a full-wave rectifier, the current in each of the diodes flows for : 1

- (A) Complete cycle of the input signal
(B) Half cycle of the input signal
(C) Less than half cycle of the input signal
(D) Only for the positive half cycle of the input signal

Ans: (B) Half cycle of the input signal

(iii) In a full-wave rectifier :

1

(A) Both diodes are forward biased at the same time.

(B) Both diodes are reverse biased at the same time.

(C) One is forward biased and the other is reverse biased at the same time.

(D) Both are forward biased in the first half of the cycle and reverse biased in the second half of the cycle.

Ans: (C) One is forward biased, and the other is reverse biased at the same time.

(iv) (a) An alternating voltage of frequency of 50 Hz is applied to a half-wave rectifier. Then the ripple frequency of the output will be :

1

(A) 100 Hz

(B) 50 Hz

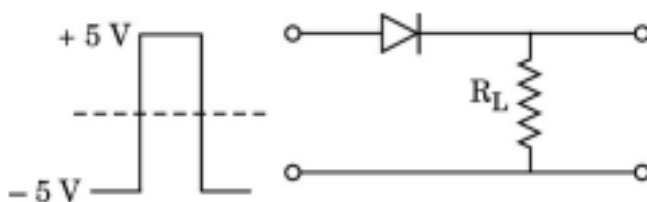
(C) 25 Hz

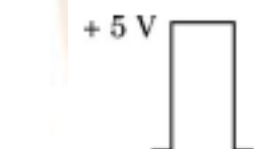
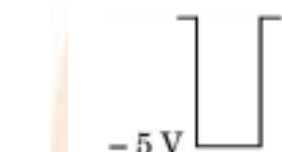
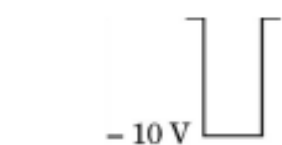
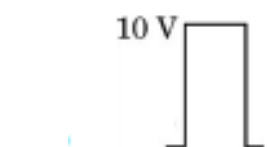
(D) 150 Hz

Ans: (B) 50 Hz

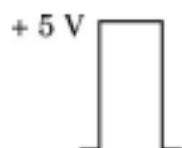
OR

(b) A signal, as shown in the figure, is applied to a p-n junction diode. Identify the output across resistance R_L :





(D)



Ans: (D)

SECTION E

31. (a) (i) Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum?

Ans: The potential energy U of an electric dipole in a uniform electric field is given by:

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

Where:

- $\vec{p}$ is the dipole moment
- $\vec{E}$ is the electric field
- θ is the angle between $\vec{p}$ and $\vec{E}$.

Maximum and Minimum Potential Energy:

I. Maximum Potential Energy:

$$U_{\max} = -pE \cos 180^\circ = pE, \text{ this occurs when } \theta = 180^\circ.$$

II. Minimum Potential Energy:

$$U_{\min} = -pE \cos 0^\circ = -pE, \text{ this occurs when } \theta = 0^\circ.$$

(ii) An electric dipole consists of point charges -1.0 pC and $+1.0 \text{ pC}$ located at $(0, 0)$ and $(3 \text{ mm}, 4 \text{ mm})$ respectively in $x-y$ plane. An electric field $\vec{E} = \left(\frac{1000 \text{ V}}{\text{m}}\right)\hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole. **5**

Ans: Torque on a Dipole: The torque $\vec{\tau}$ on a dipole in a uniform electric field is given by:

$$\vec{\tau} = \vec{p} \times \vec{E}$$

First, calculate the dipole moment $\vec{p}$:

$p = q \cdot d$, where $q = 1.0 \times 10^{-12} \text{ C}$ and $d = \sqrt{(3 \times 10^{-3})^2 + (4 \times 10^{-3})^2} = 5 \times 10^{-3} \text{ m}$ then:

$$p = (1.0 \times 10^{-12} \text{ C}) \cdot (5 \times 10^{-3} \text{ m}) = 5 \times 10^{-15} \text{ C} \cdot \text{m}$$

The dipole moment vector $\vec{p}$ points from negative to positive charge, in the direction $\hat{i} \cos \theta + \hat{j} \sin \theta$ with θ the angle with x-axis. For coordinates given:

$$\theta = \tan^{-1} \left(\frac{4}{3} \right) = 53.13^\circ$$

Thus,

$$\vec{p} = 5 \times 10^{-15} (\cos 53.13^\circ \hat{i} + \sin 53.13^\circ \hat{j}) = 5 \times 10^{-15} \left(\frac{3}{5} \hat{i} + \frac{4}{5} \hat{j} \right) = 3 \times 10^{-15} \hat{i} + 4 \times 10^{-15} \hat{j}$$

This gives the torque:

$$\vec{\tau} = \vec{p} \times \vec{E} = (3 \times 10^{-15} \hat{i} + 4 \times 10^{-15} \hat{j}) \times (1000 \hat{i}) = 0 \hat{i} + 0 \hat{j} - (3 \times 10^{-15} \times 1000) \hat{k} = -3 \times 10^{-12} \hat{k} \text{ Nm}$$

Required value of torque is $\vec{\tau} = -3 \times 10^{-12} \hat{k} \text{ Nm}$

OR

(b) (i) An electric dipole (dipole moment $\vec{p} = p\hat{i}$), consisting of charges $-q$ and q , separated by distance $2a$, is placed along the x-axis, with its centre at the origin. Show that the potential V , due to this dipole, at a point $x, (x \gg a)$ is equal to $\frac{1}{4\pi\epsilon_0} \cdot \frac{\vec{p} \cdot \hat{i}}{x^2}$.

Ans: Potential due to Dipole:

For a dipole along the x-axis with charges -q and q at -a and a:

$$V = V_+ - V_- = \frac{q}{4\pi\epsilon_0(x-a)} - \frac{q}{4\pi\epsilon_0(x+a)}$$

For $x \gg a$:

$$V \approx \frac{q}{4\pi\epsilon_0 x} \left(1 - \frac{a}{x}\right) - \frac{q}{4\pi\epsilon_0 x} \left(1 + \frac{a}{x}\right) = \frac{q}{4\pi\epsilon_0 x} \left(\frac{2a}{x}\right)$$

$$V \approx \frac{q \cdot 2a}{4\pi\epsilon_0 x^2} = \frac{p}{4\pi\epsilon_0 x^2}$$

Since $p = q \cdot 2a$:

$$V_x = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{x^2}$$

(ii) Two isolated metallic spheres S_1 and S_2 of radii 1 cm and 3 cm respectively are charged such that both have the same charge density $\left(\frac{2}{\pi} \times 10^{-9}\right) \text{ C/m}^2$. They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere S_1 . **5**

Ans: Charge Density:

Let σ be the charge density:

$$\sigma = \frac{Q_1}{4\pi(0.01)^2} = \frac{Q_2}{4\pi(0.03)^2}$$

Given both have same charge density:

$$\frac{Q_1}{0.0001\pi} = \frac{Q_2}{0.0009\pi}$$

So, $Q_2 = 9Q_1$

When connected by wire, they reach the same potential. Since:

$$V = \frac{Q}{4\pi\epsilon_0 R}$$

We get, $\frac{Q_1}{0.01} = \frac{Q_2}{0.03}$

This gives new charge on sphere S_1 :

$$Q_1 + Q_2 = Q_{\text{total}} \Rightarrow Q_1 + 9Q_1 = 10Q_1$$

$$Q_{S_1}' = \frac{Q_{\text{total}}}{10}$$

32. (a) (i) A resistor and a capacitor are connected in series to an ac source $v = v_m \sin \omega t$. Derive an expression for the impedance of the circuit.

Ans:

Impedance in Series RC Circuit:

Impedance Z is: $Z = \sqrt{R^2 + X_C^2}$ where $X_C = \frac{1}{\omega C}$

$$Z = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

(ii) When does an inductor act as a conductor in a circuit? Give reason for it.

Ans:

Inductor as Conductor: An inductor acts as a conductor when the frequency of the AC source is very high. At high frequencies, the reactance $X_L = \omega L$ becomes very large, leading to negligible current, and effectively the inductor behaves like a short

(iii) An electric lamp is designed to operate at 110 V dc and 11 A current. If the lamp is operated on 220 V, 50 Hz ac source with a coil in series, then find the inductance of the coil. 5

Ans:

The lamp is designed to operate at 110 V and 11 A DC. Using Ohm's Law, we can find the resistance R:

$$R = \frac{V}{I} = \frac{110\text{V}}{11\text{A}} = 10\Omega$$

Series Connection with an Inductor:

When the lamp and inductor are in series, the total voltage drop across the series combination should equal the supply voltage (220 V). The impedance Z of the series circuit (resistor R and inductor L) can be found using:

$$Z = \sqrt{R^2 + (\omega L)^2}, \text{ where } \omega = 2\pi f \text{ is the angular frequency and } f = 50 \text{ Hz.}$$

This gives, $\omega = 2\pi \times 50 \text{ Hz} = 100\pi \text{ rad/s}$

Calculate the Impedance:

The lamp requires 110 V across it, leaving:

$$V_L = \sqrt{(220\text{V})^2 - (110\text{V})^2} = \sqrt{48400\text{V}^2 - 12100\text{V}^2} = \sqrt{36300} \approx 190.5\text{V}$$

This voltage drop V_L is due to the inductor.

Calculate the Inductance:

Using the voltage drop across the inductor and the formula for impedance:

$$V_L = I \cdot \omega L$$

$$190.5\text{V} = 11\text{A} \times 100\pi \text{ rad/s} \times L$$

Solving for L:

$$L = \frac{190.5\text{V}}{11\text{A} \times 100\pi \text{ rad/s}} = \frac{190.5}{1100\pi} \approx 0.055 \text{ H} = 55 \text{ mH}$$

The inductance of the coil should be 55 mH.

32. (b) (i) Draw a labelled diagram of a step-up transformer and describe its working principle. Explain any three causes for energy losses in a real transformer.

(ii) A step-up transformer converts a low voltage into high voltage. Does it violate the principle of conservation of energy? Explain.

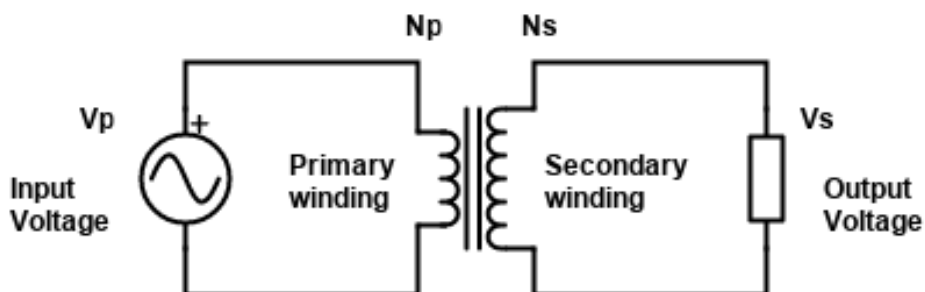
(iii) A step-up transformer has 200 and 3000 turns in its primary and secondary coils respectively. The input voltage given to the primary coil is 90 V . Calculate:

(1) The output voltage across the secondary coil

(2) The current in the primary coil if the current in the secondary coil is 2.0 A . 5

Ans:

Step-up Transformer



(i) Working Principle:

A step-up transformer increases the voltage from the primary coil to the secondary coil. It operates on the principle of electromagnetic induction. When an alternating current (AC) flows through the primary coil, it creates a varying magnetic field in the iron core. This varying magnetic field induces an electromotive force (EMF) in the secondary coil, which is proportional to the number of turns in the secondary coil.

Energy Losses in a Transformer:

1. **Copper Losses:** These occur due to the resistance of the winding wires. Energy is lost as heat when current flows through the resistance.
2. **Iron Losses:** These are caused by hysteresis and eddy currents in the iron core. Hysteresis loss is due to the repeated magnetization and demagnetization of the core, while eddy currents are circulating currents induced within the core that cause heating.
3. **Leakage Flux:** Not all the magnetic flux produced in the primary coil links with the secondary coil. Some flux leaks into the surrounding space, resulting in energy loss.

(ii) A step-up transformer does not violate the principle of conservation of energy. While it increases the voltage, it simultaneously decreases the current to maintain constant power

(ignoring losses). The power input to the primary coil ($P_p = V_p I_p$) is equal to the power output from the secondary coil ($P_s = V_s I_s$), ensuring energy conservation.

(iii) (1) Output Voltage:

Using the transformer equation:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Given, $V_p = 90\text{V}$, $N_p = 200$, and $N_s = 3000$

Substituting these values:

$$\begin{aligned} \frac{V_s}{90} &= \frac{3000}{200} \\ V_s &= 90 \times \frac{3000}{200} = 1350\text{V} \end{aligned}$$

(2) Current in Primary Coil:

Using the power equivalence (neglecting losses):

$$P_p = P_s \Rightarrow V_p I_p = V_s I_s$$

Given, $V_p = 90\text{V}$, $V_s = 1350\text{V}$, and $I_s = 2.0\text{A}$

Substituting these values:

$$\begin{aligned} 90 \times I_p &= 1350 \times 2.0 \\ I_p &= \frac{1350 \times 2.0}{90} = \frac{2700}{90} = 30\text{A} \end{aligned}$$

Therefore, the output voltage is 1350V and current in primary coil: 30 A

33. (a) (i) A ray of light passes through a triangular prism. Show graphically, how the angle of deviation varies with the angle of incidence? Hence define the angle of minimum deviation.

Ans: Graphical Representation and Explanation:

The angle of deviation (D) in a prism varies with the angle of incidence (i). As the angle of incidence increases, the angle of deviation first decreases, reaches a minimum value, and then increases again.

Angle of Minimum Deviation:

The angle of minimum deviation (D_m) is the smallest value of the angle of deviation. It occurs when the light ray passes symmetrically through the prism.

Explanation of Angle of Minimum Deviation:

The angle of minimum deviation (D_m) is the minimum value of the angle of deviation. It is defined as the angle at which light, after passing through the prism, deviates the least from its original path. At this angle, the light ray travels symmetrically through the prism, meaning that the angles of incidence and emergence are equal.

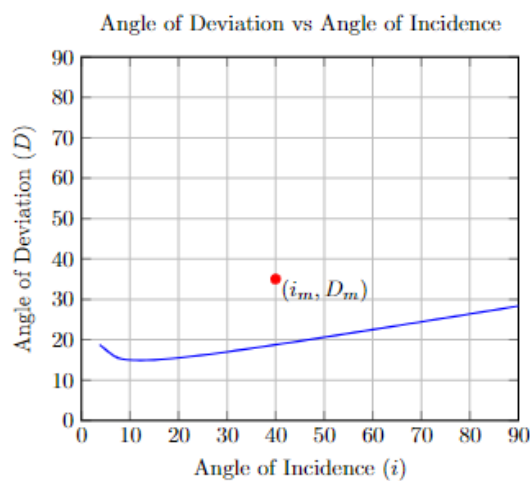


Figure 1: Graph showing the variation of the angle of deviation with the angle of incidence

(ii) A ray of light is incident normally on a refracting face of a prism of prism angle A and suffers a deviation of angle δ . Prove that the refractive index n of the material of the prism is given by $n = \frac{\sin(A + \delta)}{\sin A}$.

Ans: When light is incident normally on the first face, it enters without deviation and refracts at the second face. Using Snell's law at the second face:

$$n \sin r = \sin(A + \delta)$$

Where r is the angle of refraction inside the prism. Since the light is incident normally on the first face:

$$r = A$$

$$\text{Thus, } n \sin A = \sin(A + \delta)$$

$$n = \frac{\sin(A + \delta)}{\sin A}$$

(iii) The refractive index of the material of a prism is $\sqrt{2}$. If the refracting angle of the prism is 60° , find the

(1) Angle of minimum deviation, and

(2) Angle of incidence.

5

Ans:

(1) Angle of Minimum Deviation:

Using the formula for the minimum deviation (D_m) in a prism:

$$n = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Given $n = \sqrt{2}$, $A = 60^\circ$.

Substitute these values:

$$\sqrt{2} = \frac{\sin\left(\frac{60^\circ + D_m}{2}\right)}{\sin 30^\circ}$$

$$\sqrt{2} = 2 \sin\left(\frac{60^\circ + D_m}{2}\right)$$

$$\sin\left(\frac{60^\circ + D_m}{2}\right) = \frac{\sqrt{2}}{2} = \sin 45^\circ$$

$$\text{So, } \frac{60^\circ + D_m}{2} = 45^\circ$$

$$60^\circ + D_m = 90^\circ$$

$$D_m = 30^\circ$$

(2) Angle of Incidence:

The angle of incidence i can be found using the fact that at minimum deviation, the angle of incidence and refraction are related to the prism angle:

$$i = \frac{A + D_m}{2}$$

Given $A = 60^\circ$, $D_m = 30^\circ$:

$$i = \frac{60^\circ + 30^\circ}{2} = 45^\circ$$

Therefore, angle of minimum deviation: 30° and angle of incidence is 45° .

OR

(b) (i) State Huygens' principle. A plane wave is incident at an angle i on a reflecting surface. Construct the corresponding reflected wavefront. Using this diagram, prove that the angle of reflection is equal to the angle of incidence.

(ii) What are the coherent sources of light? Can two independent sodium lamps act like coherent sources? Explain.

(iii) A beam of light consisting of a known wavelength 520 nm and an unknown wavelength λ , used in Young's double slit experiment produces two interference patterns such that the fourth bright fringe of unknown wavelength coincides with the fifth bright fringe of known wavelength. Find the value of λ .

5

Ans:

(i) Huygens' Principle:

Huygens' principle states that every point on a wavefront serves as a source of secondary spherical wavelets. The new wavefront is the tangent to these secondary wavelets.

Constructing Reflected Wavefront:

1. Draw incident wavefronts at an angle i to the reflecting surface.
2. Use Huygens' principle to draw secondary wavelets from points where the wavefront meets the reflecting surface.
3. The tangent to these secondary wavelets forms the reflected wavefront.

Proof:

- Let i be the angle of incidence and r be the angle of reflection.
- According to Huygens' principle, the time taken for the wavefront to travel from point A to B and reflect to C must be equal.

- By geometry, this implies $\angle i = \angle r$

(ii) Coherent Sources: Coherent sources are light sources that have a constant phase difference and the same frequency.

Independent Sodium Lamps:

Two independent sodium lamps cannot act as coherent sources because their emissions are random and do not maintain a constant phase difference.

(iii) Given:

- Wavelength of known light, $\lambda_1 = 520 \text{ nm}$
- 4th bright fringe of unknown wavelength coincides with the 5th bright fringe of known wavelength.

Condition for Coincidence:

$$4\lambda = 5\lambda_1$$

$$4\lambda = 5 \times 520 \text{ nm}$$

$$\lambda = \frac{5 \times 520 \text{ nm}}{4} = 650 \text{ nm}$$

$$\therefore \lambda = 650 \text{ nm}$$